

**BDNKStar: neutron-star oscillation modes in causal first-order
viscous hydrodynamics — methods, mode census, frame
robustness, and a 3+1D engine validation**

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Abstract

We document the numerical methods implemented in BDNKSTAR, a Julia toolkit for relativistic stellar oscillations with Bemfica–Disconzi–Noronha–Kovtun (BDNK) causal first-order viscous hydrodynamics, and report the results of a complete mode census. The one-dimensional frequency-domain solvers cover radial and non-radial Cowling f/p towers, buoyancy (g) modes of stratified stars, interface (i) modes at a first-order phase transition, full-GR polar quasi-normal modes with gravitational-wave damping, axial viscous w -modes, tidal deformability, moment of inertia, and r -modes, across seven equations of state. The viscous sector is recomputed in the three BDNK hydrodynamic frames of Clarisse *et al.*: the repository reproduces their published characteristic speeds $c_+ = 1.00/0.85/0.74$ to 10^{-16} , 3×10^{-4} and 2×10^{-3} from (a_1, a_2) alone; hydrodynamic observables are frame-invariant to round-off in flat space and to 0.34% for the stellar axial w -mode, while the non-hydrodynamic gaps move by exactly the frame ratios $4.00\times$ and $7.00\times$. A structural result is that a_1 (the energy-frame time τ_ε) is inert in the axial sector and, empirically, in the 3+1D polar sector as well. Searching for EOS-independent combinations of modes, every two-mode ratio fails across a mass sequence; the $\sqrt{\ell/2}$ scaling of the f -mode ℓ -ladder is shown to be a fixed-mass artifact and the ladder relation an algebraic identity, while a four-observable pure-spectral relation among $(f_f, \tau_f, f_w, \tau_w)$ survives at 0.70% leave-one-EOS-out scatter and a new w -Love relation at 0.38%. The relativistic Cowling approximation is shown to break the f -Love relation (1.81% vs 0.18%). Finally the three 3+1D Cartesian engines are validated against the 1D solvers: the linear Cowling engine reproduces the f -mode to $< 0.1\%$ for smooth barotropes but excites a growing surface mode for crusted EOS; the BDNK engine gives a frame-invariant viscous damping rate converging at third order to 96–97% of the Navier–Stokes dissipation integral and shows the linear stability bound $\tau_Q w_0 c_s^2 > \frac{4}{3}\eta$ to be sufficient but not necessary; the nonlinear DG engine represents a non-axisymmetric $m = 2$ perturbation and bounds the grid’s violation of the m -degeneracy at 10^{-4} . All three 3D engines freeze the metric, so gravitational-wave damping, tidal deformability and the w -modes are structurally out of their reach. We keep every retracted claim and every engine defect in the record.

CONTENTS

I. Scope and conventions	4
II. Background stars	5
A. Tolman–Oppenheimer–Volkoff integration	5
B. Equations of state	5
III. One-dimensional perturbation solvers	6
A. Radial Cowling modes	6
B. Non-radial Cowling f/p modes	6
C. Gravity modes	7
D. Interface modes at a first-order phase transition	7
E. Full-GR polar quasi-normal modes	8
F. Axial viscous w -modes	8
G. Tidal deformability, moment of inertia, r -modes	9
IV. BDNK causal viscous hydrodynamics	9
A. Constitutive relations and hydrodynamic frames	9
B. The Clarisse frames and their non-conformal generalisation	9
C. Flat-space dispersion and characteristic speeds	10
D. Polar BDNK operator (frequency domain)	10
V. The 3+1D Cartesian engines	10
A. <code>CowlingEvolve3D</code> : linearized, cut-cell surface	11
B. <code>CowlingBDNK3D</code> : full-frame causal viscous	11
C. <code>DGCart3D</code> : nonlinear nodal RKDG	11
D. Frequency extraction	12
VI. Results I: mode census across equations of state	12
VII. Results II: the three BDNK frames	15
VIII. Results III: EOS-independent combinations of modes	17
IX. Results IV: the 3+1D engine census	18

X. Retractions and defects	21
A. Claims withdrawn during verification	21
B. Defects still present in <code>src/</code>	21
XI. Conclusions	22
Acknowledgments	23
References	23

I. SCOPE AND CONVENTIONS

Throughout, $G = c = 1$. Two geometric length conventions coexist in the code and were the source of one bug found in this work: `ShumPolytrope(100.0)` (the $\Gamma = 2$, $K = 100$ polytrope) is *solar-mass*-geometric, so that its central density $\varepsilon_c = 0.00128 + 100 \cdot 0.00128^2$ gives the standard benchmark star $M = 1.4002 M_\odot$, $R = 14.154$ km; the piecewise polytropes and `PolytropeEnergy` are *kilometre*-geometric. Frequencies are converted with $\text{kHz} \rightarrow \text{km}$ and the appropriate unit length L_{unit} ; omitting the L_{unit} argument silently lowers km-geometric frequencies by 1.4766 (this exact bug exists at `test/test_universal_relations.jl:80`). Compactness is $C = 1.476625 M/R$ with M in $M_\odot$ and R in km; every data file produced here satisfies this to $< 10^{-5}$.

Three verification conventions governed the whole study and are stated once here:

1. A quantity flat at a single mass is not universal; universality is tested along a mass sequence and across EOS simultaneously.
2. Identical digits across a swept parameter mean the parameter is not being forwarded. This failure mode occurred three separate times (an unforwarded $\hat{\tau}$, a defaulted-off junction switch, and a Julia parse of `2f1` as the `Float32` literal `20.0`).
3. A time-domain frequency from a record of length T has spectral resolution $\Delta f = 1/T$; any claimed effect smaller than Δf rests on parametric peak estimation and is quoted as such, never as a resolved line.

II. BACKGROUND STARS

A. Tolman–Oppenheimer–Volkoff integration

The static background is

$$m' = 4\pi r^2 \varepsilon, \quad \nu' = \frac{2(m + 4\pi r^3 p)}{r(r - 2m)}, \quad p' = -\frac{(\varepsilon + p)(m + 4\pi r^3 p)}{r(r - 2m)}, \quad (1)$$

integrated outward by classical RK4 from a Taylor-seeded regular centre, $m(h) = \frac{4\pi}{3}\varepsilon_c h^3$, $p(h) = p_c - 2\pi(\varepsilon_c + p_c)(\varepsilon_c/3 + p_c)h^2$. The surface is located by linear interpolation of the last step at $p \rightarrow 0$, and ν is shifted so that $e^{\nu(R)} = 1 - 2M/R$. The metric is $ds^2 = -e^\nu dt^2 + e^\lambda dr^2 + r^2 d\Omega^2$ with ν, λ the *full* exponents, so $\Phi' = \nu'/2$ is the local gravity and $\alpha = e^{\nu/2}$ the lapse. The step is $h = 2 \times 10^{-4}$ (production) or 5×10^{-5} (radial modes); the interface-mode TOV steps *through* a density jump, which caps its convergence at first order in h (Sec. III D).

B. Equations of state

Nine EOS families were exercised (Table I):

Shum polytrope: $p = K\rho^2$, $\varepsilon = \rho + p$, $K = 100$ ($M_\odot$ -geometric); the Font–Stergioulas–Kokkotas benchmark star.

PolytropeEnergy: $p = \kappa\varepsilon^{1+1/n}$; used at $\Gamma = 2, 2.5, 3$ as outlier controls, and at $(\kappa, n) = (100, 1)$ as the Bussi eres EOS1 axial benchmark star.

Read *et al.* piecewise polytropes: SLy, APR4, H4, MS1 from Table III of Ref. [4]: a fixed four-piece SLy crust joined to three high-density segments $p = K_i \rho^{\Gamma_i}$ at $\rho_1 = 10^{14.7}$, $\rho_2 = 10^{15} \text{ g cm}^{-3}$, with $\varepsilon_i = (1 + a_i)\rho + K_i \rho^{\Gamma_i}/(\Gamma_i - 1)$ and a_i fixed by continuity. The curve is tabulated into a monotone cubic-Hermite **TabulatedBarotrope**. The **package default of $N = 2000$ table nodes is unsafe**: it puts a silent 6% error in SLy’s radial fundamental (4.584 vs 4.328 kHz at $N = 8000$). All production runs used $N = 8000$. Backgrounds reproduce the published M_{max} and $R_{1.4}$ to 0.03–0.9%.

Tabulated Shum: `tabulate(ShumPolytrope, 1e-11, 1e-1, 4000)`, exercising the interpolation path; agrees with the analytic Shum f/p tower to nine significant figures.

The lower table bound is load-bearing: $e_{10} = 10^{-10}$ returns a *spurious fundamental* below the true one.

Stratified ideal gas: `solve_tov_idealgas`(Γ , Γ_{struct}): hydrostatic structure follows index Γ_{struct} while perturbations are adiabatic with index Γ ; $\Gamma \neq \Gamma_{\text{struct}}$ gives $N^2 \neq 0$. The only family supporting g -modes.

Phase-transition hybrid: $\varepsilon(p) = \varepsilon_{\text{poly}}(p) + \Delta\varepsilon \Theta(p - p_t)$ on the $\Gamma = 2$ base: a genuine density discontinuity at constant pressure, which the continuous piecewise polytropes cannot represent. Lives in the shooting script, not yet in `src/eos/`.

III. ONE-DIMENSIONAL PERTURBATION SOLVERS

A. Radial Cowling modes

The Lagrangian displacement obeys $A_2\xi'' + A_1\xi' + A_0\xi = \omega^2\xi$ with $A_{0,1,2}$ explicit in $(m, p, \varepsilon, \nu, c_s, c'_s)$; second-order central differences on a uniform grid, $\xi(0) = 0$, $\Delta p(R) = 0$, solved as a matrix eigenproblem. The operator needs $d^2p/d\varepsilon^2$: **no d2pde2 method exists for TabulatedBarotrope**, so the radial solver dies with a `MethodError` on every tabulated EOS (a blocker still in `src/`). The atmosphere floor $\varepsilon_f = 10^{-9}\varepsilon_c$ is hardcoded and evaluates c_s there, producing `Inf/NaN` or a spurious mode depending on the table bounds.

B. Non-radial Cowling f/p modes

The McDermott–Van Horn–Scholl / Sotani (W, V) system [6],

$$\frac{dW}{dr} = \frac{1}{c_s^2} \left[\omega^2 r^2 \frac{e^\lambda}{e^\nu} V + \frac{\nu'}{2} W \right] - \ell(\ell+1) e^\lambda V, \quad \frac{dV}{dr} = \nu' V - \frac{e^\lambda}{r^2} W, \quad (2)$$

with $W \sim r^{\ell+1}$, $V \sim -r^\ell/\ell$ at the centre and $\Delta p = 0$ at the surface, $\omega^2 R^2 (e^\lambda/e^\nu) V(R) + \frac{\nu'}{2} W(R) = 0$. RK4 shooting from the centre and Brent root-finding in ω^2 on the surface mismatch, $N = 2500$ – 4000 steps. A hardcoded surface cut at $R(1 - 10^{-3})$ contributes $+0.015/ +0.032/ +0.055\%$ to $p_2/p_3/p_4$; f and p_1 are converged to $< 0.01\%$.

C. Gravity modes

The stratified (U, V) form [7, 8], $U = r^2 e^{\lambda/2} \xi_r$, $V = \delta P / (\varepsilon + P)$,

$$\frac{dU}{dr} = \frac{g}{c_s^2} U + e^{\lambda/2} \left[\frac{\ell(\ell+1)e^\nu}{\omega^2} - \frac{r^2}{c_s^2} \right] V, \quad \frac{dV}{dr} = e^{\lambda/2-\nu} \frac{\omega^2 - N^2}{r^2} U + g \Delta(c^{-2}) V, \quad (3)$$

with $g = \nu'/2$, $\Delta(c^{-2}) = 1/c_e^2 - 1/c_s^2$ and the relativistic Brunt–Väisälä frequency $N^2 = g^2 \Delta(c^{-2}) e^{\nu-\lambda}$. $\Gamma > \Gamma_{\text{struct}} \Rightarrow N^2 > 0 \Rightarrow$ a real g -tower below the f -mode; $\Gamma < \Gamma_{\text{struct}}$ gives $N^2 < 0$ and imaginary frequencies (convective instability), both verified. Two defects: the mode classifier at `GravityModes.jl:324` uses `argmin(nodes)` with no tie-break and misidentifies the f -mode when two zero-node modes appear; and the default scan floor $\omega_{\text{lo}}^2 = 10^{-4}$ sits *above* the entire g -tower for stratification $\delta \lesssim 0.02$, silently returning zero g -modes on a star that has them.

D. Interface modes at a first-order phase transition

Written for this work. With $Y \equiv \delta p$ and Rh related to the displacement by $\xi_r = \alpha Rh / (\omega w_0 e^\lambda)$, the shooting system is

$$\frac{dY}{dr} = \frac{\omega Rh}{\alpha} - \Phi' \frac{1 + c_s^2}{c_s^2} Y, \quad (4)$$

$$\frac{dRh}{dr} = -\frac{\omega e^\lambda}{c_s^2 \alpha} Y + \frac{\ell(\ell+1)\alpha e^\lambda}{\omega r^2} Y - \frac{d \ln g}{dr} Rh, \quad g \equiv \frac{r^2 \alpha \rho_0}{w_0 \sqrt{e^\lambda}}, \quad (5)$$

with the free-surface discriminant $D = Y + \xi_r p'_0$ normalised by $|Y| + |\xi_r p'_0|$. The *structural* fact that makes the conversion-timescale question a clean null test: since $d \ln g / dr \ni -d \ln w_0 / dr$, marching straight through the discontinuity integrates the delta function to $Rh_{\text{out}} / Rh_{\text{in}} = w_{\text{out}} / w_{\text{in}}$ exactly, i.e. it enforces ξ_r continuity by itself, while Y stays continuous because its coefficients are bounded. That is *rapid* conversion. *Slow* conversion (Lagrangian Δp continuous rather than δp) adds exactly one term at the junction,

$$Y \rightarrow Y + \xi_r \Phi' (w_{\text{out}} - w_{\text{in}}), \quad (6)$$

and the i -mode exists if and only if that term is present. The claim was challenged by a verifier as “wrong in sign” and settled numerically: integrating Rh across the resolved interface and dividing by a smooth no-jump control gives $0.824804 \rightarrow 0.819238 \rightarrow 0.816287$ as the half-width halves, first order, extrapolating to 0.8133 against the prediction 0.81342

(0.01%); the opposite sign would give 1.2294. The junction skip width ϵ and the background step h are both first-order error sources ($\Delta f_i \propto \epsilon$ exactly, as expected from dropping a shell of thickness 2ϵ), extrapolating to $f_i(\ell = 2, \Delta\epsilon = 2 \times 10^{-4}) = 0.93424 \text{ kHz}$ at $\sim 0.01\%$. Two bugs were found and are recorded because both faked a *wrong-but-plausible* answer: the junction branch was keyed to a caller-supplied kwarg that defaulted off (every early “*i*-mode” was a plain star, giving 59–178 Hz, $10\times$ below the published band), and $w_{\text{in/out}}$ were read by linear interpolation of the background, which smears the jump over one cell. Stars beyond the Seidov limit $\Delta\epsilon_S = \epsilon_{\text{out}}/2 + 3p_t/2$ are radially unstable and are flagged.

E. Full-GR polar quasi-normal modes

The Lindblom–Detweiler even-parity system [9] in the numerically stable variables $\{H_1, K, W, X\}$ as written by Krüger, Gaertig & Kokkotas [10]: two centre-regular solutions combined to satisfy $X(R) = 0$; at the surface (H_1, K) is mapped to the Zerilli function, integrated in vacuum to a matching radius a ; the purely-outgoing condition is imposed analytically via the Chandrasekhar–Detweiler transform of the Regge–Wheeler Leaver continued fraction (isospectrality); the QNM condition is the vanishing Wronskian $Z_{\text{in}}Z'_{\text{up}} - Z_{\text{up}}Z'_{\text{in}} = 0$, root-found by a complex secant. Frequencies are insensitive to a ($< 2 \times 10^{-6}$) but τ_f moves by 33–35% between $a/R = 3$ and 30 and the method branch-jumps to the w -mode by $a/R = 50$; production uses $a/R = 12$ and quotes τ only with a stated.

F. Axial viscous w -modes

A faithful port of Bussi eres *et al.* [11]: coupled axial wave equations for the metric ψ and the fluid Z with the Regge–Wheeler potential and BDNK couplings, two-seed RK4 interior shooting with the surface regularity condition, exterior vacuum RW integration, Leaver “up” continued fraction, and $\Delta(\omega) = \psi'_{\text{in}}/\psi_{\text{in}} - \psi'_{\text{up}}/\psi_{\text{up}} = 0$. Viscosity parametrizations A: $\eta = \hat{\eta}(\rho + p)L_0c_s^2$, $\tau = \hat{\tau}L_0\hat{\eta}$; B: $\eta = \hat{\eta}pL_0$, $\tau = \hat{\tau}L_0p/\rho$. Reproduces Table II A1 of Ref. [11] to $+0.0002\%$ in f and -0.035% in τ ; a single solve takes 0.05–1 s. A structural observation (Sec. VII): the solver assigns the energy-frame coefficient θ at line 291 and never reads it.

G. Tidal deformability, moment of inertia, r -modes

k_2 from the Riccati equation $ry' + y^2 + yF + r^2Q = 0$, $y(0) = 2$, with the Hinderer closed form [12, 13] and $\Lambda = \frac{2}{3}k_2C^{-5}$. The moment of inertia from Hartle's frame-drag equation $\frac{d}{dr}[r^4j'\varpi] + 4r^3j'\varpi = 0$, $j = e^{-(\nu+\lambda)/2}$ [14], with $I = J/\Omega$ and $\bar{I} = I/M^3$. r -modes from Lindblom–Owen–Morsink [15]: $\sigma = \frac{2}{3}\Omega$, $\omega = -\frac{4}{3}\Omega$, with $1/\tau = 1/\tau_{\text{GW}} + 1/\tau_{sv} + 1/\tau_{bv}$; the Newtonian $n = 1$ Lane–Emden integral $\tilde{J} = 1.635270 \times 10^{-2}$ reproduces LOM98's 1.635×10^{-2} (+0.02%). One calibrated constant absorbs the $O(1)$ $\delta\rho$ eigenfunction normalisation of the bulk-viscosity integral.

IV. BDNK CAUSAL VISCOUS HYDRODYNAMICS

A. Constitutive relations and hydrodynamic frames

The general-frame stress tensor [2]

$$T^{\mu\nu} = (\varepsilon + \mathcal{A})u^\mu u^\nu + (P + \Pi)\Delta^{\mu\nu} - 2\eta\sigma^{\mu\nu} + u^\mu \mathcal{Q}^\nu + u^\nu \mathcal{Q}^\mu, \quad (7)$$

$$\mathcal{A} = \tau_\varepsilon[D\varepsilon + (\varepsilon + P)\theta], \quad \Pi = -\zeta\theta + \tau_P[D\varepsilon + (\varepsilon + P)\theta], \quad (8)$$

$$\mathcal{Q}^\nu = \tau_Q[(\varepsilon + P)Du^\nu + c_s^2\Delta^{\nu\lambda}\partial_\lambda\varepsilon], \quad (9)$$

with $D = u \cdot \nabla$, $\theta = \nabla \cdot u$. The relaxation times $(\tau_\varepsilon, \tau_P, \tau_Q)$ are the frame; physical predictions must not depend on them at first order in gradients, while the non-hydrodynamic (gapped) modes do.

B. The Clarisse frames and their non-conformal generalisation

Clarisse *et al.* [1] parametrize $\tau_\varepsilon = a_1(1/T)(\eta/s)$, $\tau_Q = a_2(1/T)(\eta/s)$ under $a_1 \geq 4$, $a_2 \geq 3a_1/(a_1 - 1)$, and use (Table VII) $F_1 = (25/4, 25/7)$, $F_2 = (25/2, 25/3)$, $F_3 = (25, 25)$. F_1 saturates the second bound exactly, $3(25/4)/(21/4) = 25/7$, which is why its c_+ is exactly 1: it is the marginally causal frame, and any stellar result in which F_1 looks cleaner than F_2, F_3 should be treated as suspect. At zero chemical potential $w = \varepsilon + P = sT$, so $(1/T)(\eta/s) = \eta/w$ and the stellar map is

$$\tau_\varepsilon = a_1 \eta/w_0(r), \quad \tau_Q = a_2 \eta/w_0(r), \quad (10)$$

position-dependent inside a star. Under this map the repository’s own linear stability bound for the polar sound sector, $\tau_Q w_0 c_s^2 > \frac{4}{3}\eta$, reduces to the η -independent condition $a_2 c_s^2 > 4/3$.

C. Flat-space dispersion and characteristic speeds

The sound-channel quartic of Kovtun [3] in the general frame $(\varepsilon_1, \varepsilon_2, \pi_1, \pi_2, \theta)$ is solved by companion-matrix roots; the Clarisse frames map as $\theta = \tau_Q w$, $\varepsilon_1 = \tau_\varepsilon w / c_s^2$, $\varepsilon_2 = \pi_1 = c_s^2 \varepsilon_1$, $\pi_2 = c_s^4 \varepsilon_1 - \zeta$ (the last being Kovtun’s footnote-8 compensation that pins the physical bulk viscosity across frames). The large- k biquadratic (Kovtun Eq. 4.18) gives the characteristic speeds. The separate Keeble–Redondo–Yuste biquadratic in `Causality.jl` is a different characteristic system (it lacks the π_1, ε_2 cross terms) and does *not* reproduce the Clarisse table under any feeding; it is retained for cross-reference only.

D. Polar BDNK operator (frequency domain)

The linearised polar BDNK–Cowling system reduced to one radial dimension for fixed ℓ , five fields $(\hat{\rho}, \delta \hat{S}_r, \delta \hat{S}_\theta, \hat{u}^r, \hat{u}^\theta)$, plus a sixth (the Lagrangian displacement ξ) when stratified; QNMs are the eigenvalues of the resulting matrix, tracked by complex continuation in $\hat{\eta}$. Transport channels: relaxation $\tau_R = \hat{\tau} + c_\tau \hat{\eta}$ (recovery denominator $\tau_R w_0 - \eta$), reactive conduction κ_Q , and momentum shear ν_{mom} (the genuine damper). A corrected free-surface boundary condition ($\Delta p = 0 \Rightarrow \partial_t \delta \rho = -\delta v^r p'_0 / (c_s^2 h_0)$) moved the f -mode from +8.91% to +0.18% of the shooting value. A verifier found that the velocity recovery omits the lapse multiplying the whole bracket, so the faithful relaxation slot is $\tau_R = \tau_Q / \alpha$; the polar *damping* rates from this operator were withheld on that basis (Sec. VII).

V. THE 3+1D CARTESIAN ENGINES

All three engines freeze the metric (Cowling). This is a structural limit, not a resolution one: there is no gravitational radiation and no metric perturbation, so w -modes, GW damping times, Λ , and hence f -Love and w -Love cannot exist in them; neither can g , i , or r modes (no stratification, no discontinuity, no rotation).

A. CowlingEvolve3D: linearized, cut-cell surface

Evolves $(\delta\varepsilon, \delta S_i)$ with $\delta S_i = (\varepsilon_0 + p_0)\delta v_i$ so that $(\varepsilon_0 + p_0)$ cancels from every RHS and the $1/w_0$ surface blow-up disappears:

$$\partial_t \delta\varepsilon = -\frac{1}{\sqrt{\gamma}} \partial_i [\alpha \sqrt{\gamma} \delta S^i] - \alpha \Phi'(n_i \delta S^i), \quad \partial_t \delta S_i = -\alpha [\partial_i \delta p + \Phi' n_i (\delta p + \delta\varepsilon)], \quad (11)$$

$\gamma^{ij} = \delta_{ij} + (e^{-2\lambda} - 1)n_i n_j$, second-order central differences, Kreiss–Oliger dissipation σ_{KO} , RK4. The surface is treated by cut cells: both the volume fraction κ and the six face apertures reduce to one-dimensional quadratures of the same disc–rectangle area, verified against the analytic sphere to 10^{-7} in volume and $4\pi R^2$ at second order in closure. Seeds and diagnostics are $\ell = 2$, $m = 0$ only ($2Z^2 - X^2 - Y^2$). **The default cut=false is wrong by -6.95% and does not converge away between $N = 32$ and 40 , from a run with no visible symptom.**

B. CowlingBDNK3D: full-frame causal viscous

Linearised BDNK with the frame times kept, so the dissipative corrections enter the conserved densities and the system is hyperbolic. The primitives are recovered from

$$\delta E = \delta\varepsilon + \tau_\varepsilon \left(\frac{1}{\alpha} \partial_t \delta\varepsilon + w_0 \theta \right), \quad \delta S^i = w_0 \delta v^i + (\tau_Q w_0 - \eta) \frac{1}{\alpha} \partial_t \delta v^i + \tau_Q c_s^2 \gamma^{il} \partial_l \delta\varepsilon, \quad (12)$$

with the covariant expansion $\theta = \partial_i \delta v^i + (\Phi' + A'/A)(n \cdot \delta v)$ and the closure $\tau_P = c_s^2 \tau_\varepsilon$ forced by $\Pi = c_s^2 \mathcal{A}$ (using $\tau_P = \tau_\varepsilon$ reopens a documented primitive-feedback instability). The recovery denominator $\tau_Q w_0 - \eta$ is the causality margin. The viscous damping is measured by the *difference protocol*: $\gamma_{\text{visc}} = \gamma(\hat{\eta}) - \gamma(0)$ at identical grid and frame, which cancels the numerical floor; at $N = 40$ that floor is still 52% of γ_{visc} , so the protocol is mandatory.

C. DGCart3D: nonlinear nodal RKDG

Full 3D tensor-product nodal Runge–Kutta discontinuous Galerkin Valencia GRHydro on the frozen metric: densitized conserved set $D = \sqrt{\gamma} \rho W$, $S_i = \sqrt{\gamma} \rho h W^2 v_i$, $\tau = \sqrt{\gamma} (\rho h W^2 - p - \rho W)$, directional Rusanov sweeps, SSP-RK3, a per-element TVB troubled-cell / Zhang–Shu positivity limiter, an atmosphere, and the well-balanced lake-at-rest device (the static background’s DG RHS is stored and subtracted). Octant grid with reflecting symmetry and

a full 3-momentum, so $\text{Re } Y_{22} \propto n_x^2 - n_y^2$ can be represented; `dgcart3d_quadrupole_m2` is the $m \neq 0$ detector. Two facts govern everything it produced: the octant’s three *separate* reflections admit only the E_g doublet $\{Y_{20}, \text{Re } Y_{22}\}$ of the $\ell = 2$ quintet, so the T_{2g} triplet (all of $m = 1$) is structurally absent; and **its background is not static** — the well-balanced subtraction is exact but `_limit!` is not, so ρ_c erodes -10% by $t = 600 M_\odot$ (4.6 f -periods) and -31 to -43% by $t = 3000$.

D. Frequency extraction

Time series are analysed by a parabolic-vertex-refined periodogram and an independent matrix-pencil (SVD, 14–16 poles) estimator that must agree; every run is gated on finiteness and on the end/start envelope ratio before any transform, and a record whose clean window is below 10 periods is voided. At affordable resolution the usable record is 23 periods ($\Delta f/f = 4.4\%$) for the linear and BDNK engines and 4.6 periods ($\Delta f/f = 21.6\%$) for the DG engine — capped not by cost but by the background erosion above. *Every sub-percent 3D figure in this paper is a parametric estimate, not a resolved line.*

VI. RESULTS I: MODE CENSUS ACROSS EQUATIONS OF STATE

TABLE I. Coverage. ✓ computed; — unsupported by construction; × blocked by a defect in `src/`.

* radial spectra of the piecewise polytropes are not resolution-independent (closure of $d^2p/d\varepsilon^2$; APR4 fundamental moves $4.97 \rightarrow 4.52 \rightarrow 4.58$ kHz as the table refines).

EOS	radial	f/p	g	i	axial w	polar GR	r
ShumPolytrope(100)	✓	✓	—	—	✓	✓	✓
PolytropeEnergy	✓	✓	—	—	✓	✓	✓
SLy, APR4, H4, MS1	✓*	✓	—	—	✓	✓	✓
TabulatedBarotrope	×	✓	—	—	✓	✓	✓
IdealGasStar ($\Gamma \neq \Gamma_{\text{struct}}$)	×	—	✓	—	✓	—	—
phase-transition hybrid	×	✓	—	✓	✓	—	—

The i -mode passes four independent tests. (i) Exactly one new eigenvalue appears under

TABLE II. Non-radial Cowling f/p tower, $M = 1.4 M_\odot$ unless stated, kHz. Benchmark: Font–Stergioulas–Kokkotas as tabulated in Ref. [5] Table VI. The tabulated path reproduces the analytic to nine figures. p_2 – p_4 of the piecewise polytropes wobble non-monotonically at 0.1–0.3% under resolution change.

EOS	R [km]	$f_{\ell=2}$	p_1	p_2	p_3	p_4	$f_{\ell=3}$	$f_{\ell=4}$
ShumPolytrope	14.154	1.8829	4.1067	6.0311	7.8694	9.6709	2.2565	2.5666
benchmark	14.15	1.8825	4.1060	6.0298	7.8670	9.6663	—	—
SLy	11.688	2.4102	6.878	9.611	10.769	12.308	2.9462	3.3954
APR4	11.302	2.5082	7.323	10.263	11.500	13.104	3.0700	3.5403
H4	13.930	1.9458	5.493	7.291	8.214	9.716	2.3722	2.7292
MS1	14.904	1.7776	5.320	6.509	7.678	9.221	2.1701	2.4990

TABLE III. Axial w -modes (full GR, $\ell = 2$): inviscid, and viscous at $\eta_c = 10^{31}$ cgs. Bussi eres Table II A1 is reproduced to +0.0002% (f) and -0.035% (τ).

EOS	f inv. [kHz]	τ inv. [μ s]	f visc. [kHz]	τ visc. [μ s]
PolytropeEnergy(100, $n=1$)	10.501355	29.533755	10.089819	30.874839
Bussi�eres A1	10.50135	29.54	10.0898	30.8857
ShumPolytrope	7.087953	28.155033	6.564552	32.022022
PolytropeEnergy(5000, $n=2/3$)	8.892622	30.892688	8.377897	33.182050
SLy	8.036591	29.287289	7.397014	32.731680
APR4	8.211825	29.597018	7.581011	32.906530
H4	7.002713	28.620787	6.179403	34.529500
MS1	6.559351	28.766736	5.8378	34.6518
IdealGasStar (stratified)	7.087953	28.155033	6.568516	31.997322
isentropic $\Gamma = 5/3$	4.017675	32.345819	did not converge	
phase transition	7.178252	25.284205	ladder stalled at 10^{30}	

the slow map and none under the rapid map at every parameter point. (ii) $\omega_i^2/[g(r_t)k_h A]$ with $k_h = \sqrt{\ell(\ell+1)}/r_t$ is 0.8965, 0.8966, 0.8962 for $\Delta\varepsilon = 5 \times 10^{-5}, 10^{-4}, 2 \times 10^{-4}$ — linear in $\Delta\varepsilon$ to 0.02% over a factor of four — but this 0.896 is *not* a universal constant: across a wider

TABLE IV. Polar QNM beyond Cowling ($a/R = 12$) and the Cowling overestimate on the same stars. τ carries the 33–35% matching-radius systematic.

EOS	f [kHz]	τ [s]	Cowling f	overestimate
SLy	1.931936	0.194418	2.4102	+24.8%
APR4	2.004821	0.181278	2.5082	+25.1%
H4	1.532549	0.307263	1.9458	+27.0%
MS1	1.373709	0.385505	1.7776	+29.4%
ShumPolytrope	2.323442	0.202532	—	—

TABLE V. g -modes, $\ell = 2$, stratified ideal gas, $\delta = \Gamma/\Gamma_{\text{struct}} - 1$, Hz. Scaling: $d \log g_n / d \log N_{\text{max}} = 0.981$ (theory 1) and $d \log g_n / d \log \delta = 0.481$ (theory 1/2). g_1/g_2 is constant to 0.1% only for $\delta \leq 0.05$, drifting to $\sim 1\%$ by $\delta = 0.32$.

δ	g_1	g_2	g_3
0	zero g -modes ($N^2 \equiv 0$, as required)		
0.005	275.95	208.91	167.15
0.010	388.33	293.95	235.21
0.020	543.85	411.59	329.38
0.050	835.85	632.36	506.27
0.100	1130.42	855.36	685.44
0.200	1473.37	1117.87	897.97
0.320	1707.78	1303.66	1050.69

interface sweep it drifts as a finite-wavelength/sphericity correction. (iii) $\omega_i^2(\ell)/\omega_i^2(2) = 1.504, 1.984, 2.453$ for $\ell = 3, 4, 5$ tracks $\sqrt{\ell(\ell+1)/6} = 1.414, 1.826, 2.236$ (the interfacial law $\omega^2 \propto k_h$) and rejects $\omega^2 \propto k_h^2$ (which would give 2.00, 3.33, 5.00). (iv) $|\xi_r|$ peaks at $r/R = 0.4730 = r_t/R$. Raising $\Delta\epsilon$ walks the mode up through the spectrum into an avoided crossing with p_1 (a 3.252/3.379 kHz pair near $\Delta\epsilon = 2 \times 10^{-3}$).

TABLE VI. Interface modes, $\ell = 2$, $\Delta\epsilon = 10^{-4}$, interface-position sweep on the $\Gamma = 2$ base (this session’s regeneration). The i -mode is absent in every rapid-conversion run. The row at $r_t/R = 0.686$ returned two slow roots without rapid counterparts and is left unidentified rather than resolved by choice.

p_t/p_c	r_t/R	$M [M_\odot]$	Atwood A	f_i [kHz]	f [kHz]	p_1 [kHz]
0.10	0.686	1.2146	0.1026	(ambiguous, 2 roots)	1.9205	4.2952
0.18	0.602	1.2470	0.0767	0.694149	1.8938	4.1643
0.25	0.547	1.2710	0.0648	0.674687	1.8803	4.1175
0.35	0.481	1.3000	0.0542	0.648432	1.8697	4.0873
0.45	0.423	1.3241	0.0473	0.625138	1.8652	4.0756
0.55	0.370	1.3443	0.0424	0.604614	1.8644	4.0730
0.70	0.289	1.3689	0.0370	0.578083	1.8675	4.0780
0.85	0.197	1.3875	0.0331	0.555569	1.8739	4.0897

TABLE VII. The Clarisse frames [1] and the validation gate: $c_\pm$ computed here from (a_1, a_2) alone via the Kovtun large- k biquadratic, conformal $c_s^2 = 1/3$. Differences from the published column are 1.1×10^{-16} , 2.8×10^{-4} , 1.9×10^{-3} — two-decimal rounding.

frame	a_1	a_2	c_+ publ.	c_+ here	c_- here	bound $a_2 \geq 3a_1/(a_1 - 1)$
F_1	25/4	25/7	1.00	1.000000	0.200000	saturated exactly
F_2	25/2	25/3	0.85	0.849719	0.323488	satisfied
F_3	25	25	0.74	0.741915	0.411779	satisfied

VII. RESULTS II: THE THREE BDNK FRAMES

Two cautions on the mapping. `conformal_frame.PMP` in `Transport.jl` uses $\lambda = (25/7)\eta$, $\chi = (25/4)\eta$, i.e. it *is* F_1 , but its slot assignment is reversed relative to the a_1/a_2 roles and must not be cited as independent corroboration. And $c_+(F_1) = 1$ only at $c_s^2 = 1/3$; at $c_s^2 = 0.25$ the same (a_1, a_2) gives a superluminal $c_+ = 1.053$.

Three results stand out from Table VIII.

The control is real. A deliberate $40\times$ frame perturbation moves both flat-space gaps by

TABLE VIII. Frame robustness at fixed physical transport. Physical quantities must be flat; non-hydrodynamic ones must move. Sound channel: Kovtun quartic, $\eta = 0.30$; conformal $\zeta = 0$, NS-like $\zeta = 0.015$. Axial: $\eta_c = 10^{31}$ cgs, parametrization A. Polar: $\hat{\eta} = 0.03$, $N_r = 140$, corrected free-surface BC, percentages against 1.8829 kHz.

sector	quantity	F_1	F_2	F_3	spread
sound $c_s^2=1/3$	$\text{Re } \omega/k$	0.577350269190	0.577350269190	0.577350269190	0.0%
	Γ	0.4000000	0.4000000	0.4000000	0.026% at $k = 10^{-2}$
	gap $1/\tau_\epsilon$ (gauge)	0.5333	0.2667	0.1333	128.6% (4.00 $\times$)
	gap $1/\tau_Q$ (gauge)	0.9333	0.4000	0.1333	163.6% (7.00 $\times$)
sound $c_s^2=0.25$	$\text{Re } \omega/k$	0.500000000000	0.500000000000	0.500000000000	0.0%
	Γ	0.4150000	0.4150000	0.4150000	≈ 0
axial w	f [kHz]	10.056071	10.077335	10.090470	0.34%
	$1/\tau$ [μs^{-1}]	0.0322627	0.0323921	0.0323876	0.40%
	$c_\eta(0) = 1/\sqrt{a_2}$ (gauge)	0.52915	0.34641	0.20000	91.8% ($\sqrt{7}$)
	gap $1/\tau_Q(0)$ (gauge)	0.092315	0.039563	0.013188	163.6%
polar	f , $\hat{\eta}=0.03$	1.9038 (+1.11%)	1.9105 (+1.47%)	1.9272 (+2.35%)	1.22%
	f , $\hat{\eta} = 0$	1.8870 (+0.22%)	1.8868 (+0.21%)	1.8848 (+0.10%)	0.013%
	f , $\hat{\eta}=0$, old BC	2.0121 (+6.86%)	2.0093 (+6.71%)	1.9881 (+5.58%)	0.64%
	cells violating bound	100%	60.7%	21.4%	—

exactly 40.00 $\times$ while v_s and Γ stay flat to $1.2 \times 10^{-12}\%$; an a_2 -only 112 $\times$ sweep moves only the heat-flux gap. A frame-independence claim without such a control is void.

a_1 is inert in the axial sector. $\mathcal{A} = \tau_\epsilon[u \cdot \nabla \epsilon + w \nabla \cdot u]$ is a scalar; odd-parity perturbations have $\delta \epsilon = \delta P = 0$ and $\nabla \cdot \delta u = 0$, so $\mathcal{A} \equiv 0$ and τ_ϵ never enters. The code agrees. So $F_1/F_2/F_3$ are three values of τ_Q and nothing else there, and the 0.34% residue is not an offset: it scales as $\eta^{1.99}$ while the physical viscous shift (-4.06% from the inviscid 10.501355 kHz) scales as $\eta^{0.98}$ — one full order higher in the dissipative expansion.

The polar test block is refuted. `test_frame_independence.jl` block (c) recorded a “fixed +6–7% f -mode offset present at all frames”. That reproduces exactly — under the *old* no-slip boundary condition. With the corrected free surface it collapses to +0.10–0.22%. The offset was never a solver property. The polar damping rates from this operator are withheld: with the recovery lapse restored the f -mode damping frame-spread changes by $\sim 20\times$ and the p_1 damping spread (18/25/30% at $N_r = 100/140/180$) is unresolved.

VIII. RESULTS III: EOS-INDEPENDENT COMBINATIONS OF MODES

Tested on mass sequences: 7 EOS, 88 stars, $M = 0.55\text{--}2.62 M_\odot$, $C = 0.073\text{--}0.318$, all mode families on one background grid. The pipeline gate: I-Love against the *published* Yagi–Yunes coefficients [16] with nothing fitted gives -0.171% mean, 0.293% std, 1.118% max — literature grade.

Every two-mode ratio fails (Table IX). The $\sqrt{\ell/2}$ result deserves its own paragraph because it is the canonical trap: $\omega_f(\ell)/\omega_f(2) = 1.2216, 1.4072$ at $M = 1.4$ across the four realistic EOS (spreads 0.40% , 0.63%) sits within 0.5% of $\sqrt{3/2}, \sqrt{2}$ — and drifts *monotonically with compactness*, moving $2\text{--}4\times$ more along one EOS’s mass sequence than across all EOS at fixed mass, crossing $\sqrt{\ell/2}$ at nearly the same compactness $C = 0.186\text{--}0.205$ for every EOS. $M = 1.4 M_\odot$ merely lands near that crossing. Likewise the ℓ -ladder relation $f_4/f_2 = 1.0127 (f_3/f_2)^{1.643}$, with a spectacular 0.039% leave-one-EOS-out scatter, is an algebraic identity: if $\omega_\ell \propto \ell^b$ then $\ln r_4/\ln r_3 = \ln 2/\ln 1.5 = 1.7095$ identically, the fit is a reparametrisation of the single number b (which ranges $0.44\text{--}0.53$ and drifts with C), and $b \approx 0.49$ is the known surface-wave asymptotic.

TABLE IX. Two-mode ratios: spread at $M = 1.4$ (four realistic EOS) versus across the full sequence. None survives.

ratio	fixed $M = 1.4$	across sequence
$f(\ell=3)/f(\ell=2)$	0.40%	3.95%
$f(\ell=4)/f(\ell=2)$	0.63%	6.44%
p_1/f	5.8%	33%
f_w/f_f	15.4%	86%
τ_f/τ_w	78.9%	843%

What survives. Two modes fix (M, R) EOS-independently (the Andersson–Kokkotas inversion [17]); a third and fourth are over-determined, and eliminating (M, R) leaves a relation among observables alone:

$$\ln \frac{f_w}{f_f} = \text{quad}[\ln(f_f \tau_f), \ln(f_w \tau_w)], \quad (13)$$

leave-one-EOS-out 0.702% std / 3.28% max over 88 stars against an 18.3% constant baseline ($\approx 0.3\%$ excluding the $\Gamma = 2$ control). It does *not* beat the (M, R) route at matched param-

eter count — $\text{quad}(C, \ln \bar{I}) = 0.59\%$ — its value is needing no mass or radius measurement, not accuracy. Table X gives the fitted relations. The w -Love relation is new and approximately universal; leave-one-EOS-out the worst non-outlier is 1.81% and the $\Gamma = 2$ control degrades to 5.67%. And a directly actionable negative: *the relativistic Cowling approximation breaks f -Love* — 1.814% RMS (1.204% realistic-4) against full-GR 0.181% (0.119%) — so Cowling f -modes cannot be used in an f -Love inversion.

TABLE X. Universal-relation fits, $\ln y = \sum_{k=0}^3 a_k x^k$, 88 stars. Dimensionless $M\omega$ with $\omega = 2\pi f$ in km^{-1} , M in km. RMS/max over all 7 EOS; “real.-4” the four Read *et al.* EOS refit.

relation	y	x	a_0, a_1, a_2, a_3	RMS	max	real.-4
f -Love (full GR)	$M\omega_f$	$\ln \Lambda$	$-1.6846, -0.05492, -0.017775, 4.981\text{e-}4$	0.181%	0.758%	0.035%
f -Love (Cowling)	$M\omega_f^{\text{cow}}$	$\ln \Lambda$	$-1.5736, -0.04594, -0.014385, 2.020\text{e-}4$	1.814%	6.265%	0.586%
w -Love (new)	$M\omega_w$	$\ln \Lambda$	$-0.78206, 0.040673, -0.018318, 4.874\text{e-}4$	0.379%	1.397%	0.088%
f -I	$M\omega_f$	$\ln \bar{I}$	$-0.34324, -0.93378, 0.027140, -1.506\text{e-}3$	0.338%	1.606%	0.090%
Q-Love $_f$	$\omega_f \tau_f$	$\ln \Lambda$	$8.1245, -0.37291, 0.065692, -1.973\text{e-}3$	0.994%	4.609%	0.232%
Q-Love $_w$ (new)	$\omega_w \tau_w$	$\ln \Lambda$	$2.1675, -0.52356, 0.047319, -1.827\text{e-}3$	1.042%	4.674%	0.347%
fdamp-Love	M/τ_f	$\ln \Lambda$	$-9.8091, 0.31799, -0.083466, 2.471\text{e-}3$	1.138%	5.130%	0.266%
wdamp-Love (new)	M/τ_w	$\ln \Lambda$	$-2.9496, 0.56423, -0.065636, 2.315\text{e-}3$	1.023%	5.368%	0.375%
AK98 w -freq	$R\omega_w$	C	$2.2938, -0.96552, -5.1847$ (quadratic)	1.670%	5.285%	0.596%
AK98 f -freq	f_{GR} [kHz]	$\sqrt{M/R^3}$	$0.48166, 1.79196$ (linear)	4.564%	16.9%	3.343%

IX. RESULTS IV: THE 3+1D ENGINE CENSUS

Four blocks ran and were re-run bit-for-bit by verifiers. Deviations are against the 1D value for the identical star. Voided rows are kept.

The discriminant for the crusted-EOS instability is the surface index n in $\varepsilon \propto (R - r)^n$: 0.50/0.67/1.005 (stable) versus 1.84/1.87/1.95/1.95 (unstable), monotone over 19 decades of end/start ratio. The atmosphere-sound-speed hypothesis was tested (all 77 040 atmosphere cells zeroed, SLy $N = 48$) and refuted (+0.596 vs +0.595 e-folds/period). No affordable configuration gives a converging crusted-EOS f -mode; the mechanism is open.

Converged quantities: frame spread of γ_{visc} $0.150 \rightarrow 0.130 \rightarrow 0.045\%$ at $N = 24/32/40$

TABLE XI. CowlingEvolve3D, $\ell = 2$, $m = 0$, cut-cell, 23-period records ($\Delta f/f \approx 4.4\%$). Deviation vs the 1D shooting value.

EOS	N	σ_{KO}	mode	f_{3D} [kHz]	f_{1D} [kHz]	dev / status
$\Gamma = 2$ anchor	24/32/40/48	0.01	f	1.88301/1.87560/1.87701/1.88135	1.88291	+0.005/ − 0.388/ − 0.313/ − 0.083%
	48	0.01	p_1	4.0665	4.10672	−0.98%
	48	0.02	f	1.89076	1.88291	+0.417% (σ_{KO} systematic)
	32, 40	0.004	f	1.75235, 1.75205	1.88291	−6.93/ − 6.95% (cut=false default)
$\Gamma = 2.5$	48	0.01	f, p_1	2.23448, 5.8517	2.23404, 5.9007	+0.020%, −0.83%
$\Gamma = 3$	48	0.01	f, p_1	2.40545, 7.0718	2.40739, 7.1457	−0.080%, −1.03%
SLy	32	0.01	f	2.37154	2.41928	−1.973%; Rich. −1.024%
	40, 48	0.01	f	VOID		surface mode grows
	48	0.04	f	2.31598		−4.270%, KO-biased
APR4	32	0.01	f	2.45373	2.49463	−1.640%; Rich. −0.509%
	40, 48	0.01	f	VOID		end/start 8×10^{14}
H4	32	0.01	f	1.89169	1.94832	−2.907%; Rich. −2.572%
MS1	24	0.01	f	1.69672	1.76756	−4.008% (only stable N)
	40, 48	0.04	f	1.64607, 1.64294		−6.873/ − 7.050%
crusted (4)	24, 32	0.01	p_1	4.67–6.43	5.28–7.26	−10 to −18%, unresolved

TABLE XII. CowlingBDNK3D, $\ell = 2$, $m = 0$, $\hat{\eta} = 0.03$, cut-cell, $\sigma_{\text{KO}} = 0.01$, $N = 40$, $T = 2500 M_\odot$ (22.8 periods). $\gamma_{\text{visc}} = \gamma(\hat{\eta}) - \gamma(0)$.

frame	f_{3D} pencil/pgram	f_{1D} polar	$\gamma_{\text{visc}} [10^{-3}/M_\odot]$	τ_{damp}, Q	bound violated	stable
F_1	1.87954/1.88027	1.9038	2.48251	1.984 ms, 11.72	100.0%	yes (6.4×10^{-5})
F_2	1.87973/1.88036	1.9105	2.48229	1.984 ms, 11.72	94.4% (94.9% vol.)	yes
F_3	1.88040/1.88082	1.9272	2.48139	1.985 ms, 11.73	51.3% (51.3% vol.)	yes
$\hat{\eta} = 0$	1.87681 (all)	1.887	—	—	—	yes
$F_1, F_3, N=32, 46$ per.	—	—	decay const. to 1%	—	—	12 decades

(shrinking; the 1D 14.28% was a lapse artifact and is thereby independently confirmed as such); $\gamma_{\text{visc}}(F_1) = 2.35482 \rightarrow 2.44941 \rightarrow 2.48251 \times 10^{-3}$, order ≈ 3 , Richardson $2.51 - 2.54 \times 10^{-3} = 96 - 97\%$ of the Navier–Stokes dissipation integral $0.08746\hat{\eta} = 2.6238 \times 10^{-3}$; numerical floor $\gamma(0) = 3.27/1.96/1.30 \times 10^{-3} \propto dx^{1.8}$. Not converged: the viscous frequency shift +0.40/ +0.28/ +0.15%, quoted as $< 0.2\%$. Two physics results. (i) The bound $a_2 c_s^2 > 4/3$ is *sufficient, not necessary*: F_1 violates it in every cell and is as stable as F_3 because the unstable band lies above $k_c = C/\hat{\eta} = 15.3/10.1/5.8 M_\odot^{-1}$, which the second-order stencil resolves only for $N > 353/232/135$; moved to the cheap $\hat{\eta}$ axis at $N = 24$ the criterion predicts 6/6 blow-ups in sign and 24–28% in growth rate, and F_3 destabilises *first* (k_c 2.6 $\times$ smaller) despite violating in fewer cells. (ii) $a_1 = 6.25/25/100$ at fixed a_2 gives

$f = 1.89112/1.89113/1.89123$ kHz: inert in 3D as in the axial sector.

TABLE XIII. **DGCart3D**, $\ell = 2$, $m = 0$ and 2 , octant, $p = 2$, window $[0, 600] M_\odot$ (4.6 periods, $\Delta f/f = 21.6\%$), $A = 10^{-2}$. On a non-rotating star the m -degeneracy is exact; all splitting is grid artifact.

K (nodes/axis)	$f(m=0)$	$f(m=2)$	dev vs 1D	$ \Delta f /f$	status
4 (12)	1.85758	1.49994	$-1.3/ -20.3\%$	19.25%	VOID, background collapses
5 (15)	1.47898	1.57348	$-21.5/ -16.4\%$	6.39%	coarse
6 (18)	1.61488	1.55356	$-14.2/ -17.5\%$	3.80%	usable
8 (24)	1.58933	1.55637	$-15.6/ -17.3\%$	2.07%	usable
10 (30)	1.57218	1.58496	$-16.5/ -15.8\%$	0.81%	best
6, Y_{21} seed, one run	1.56352 (q_{20})	1.56366 (q_{22})	-17.0%	0.0090%	degeneracy bound $\leq 10^{-4}$

The cross-run splitting falls as $h^{3.2}$ ($K = 4$ – 10), consistent with the scheme’s $O(h^3)$; the within-run Y_{21} bound is $90\times$ tighter and shows the grid’s genuine degeneracy violation is $\leq 10^{-4}$, the rest being run-to-run systematics. The Y_{21} seed is a projection diagnostic, not an $m = 1$ measurement: its mirror extension $|n_x n_z|$ projects onto the E_g doublet with predicted $q_{22}/q_{20} = 1.0000$, confirmed to 1.2×10^{-4} . The absolute f is -14 to -18% at every K , a pre-existing undiagnosed engine systematic common-mode between $m = 0$ and 2 . **Nonlinear amplitude dependence** (same engine, $K = 8$): every shift for $A \leq 3 \times 10^{-2}$ is at or below the 0.82% empirical floor; $A = 10^{-1}$ gives $-1.38 \pm 1.02\%$, “suggestive”, but the same point at $K = 6$ gives $-7.48 \pm 0.90\%$ — a $5.4\times$ change for a 25% refinement, the signature of limiter/Rusanov dissipation rather than fluid nonlinearity. The second harmonic is not detected, $A(2f)/A(f) < 0.09$ – 0.19 at $\xi/R = 0.41$. The linear engine reproduces exact amplitude proportionality to 10^{-14} , so the pipeline is not the source. *No nonlinear frequency shift is resolved at affordable cost.*

X. RETRACTIONS AND DEFECTS

A. Claims withdrawn during verification

- Radial spectra of the piecewise polytropes (APR4 fundamental 6.6% off; H4 2.0%); root cause the Hermite-table second derivative.
- All r -mode instability windows: two of five stars were the wrong mass (0.95 and $1.93 M_{\odot}$ labelled 1.4) and three more mislabelled. The algebraic result and the Lane–Emden $\tilde{J}$ stand.
- Polar BDNK damping frame-spreads (missing lapse).
- Viscous axial numbers for isentropic $\Gamma = 5/3$ and the phase-transition EOS at $\eta_c = 10^{31}$ (ladders never reached it).
- “ g_1/g_2 constant to 0.1% over $64\times$ in δ ” — true only for $\delta \leq 0.05$.
- Universal-relation claims about the Cowling f -mode’s deviation from AK98 being monotone in compactness; the true Cowling overestimate (+24.8–+29.4% at $M = 1.4$; +13.9 to +32.9% over the full 88-star set) is far larger than the +5.9–17.2% inferred from AK residuals.
- “The pure-spectral route is $2.5\times$ more EOS-independent than (M, R) ” — comparable, not superior, at matched parameter count.
- The ℓ -ladder relation as a universal relation (algebraic identity); the $\sqrt{\ell/2}$ scaling as anything but a fixed-mass artifact; the i -mode’s 0.896 factor as a universal constant.
- Early i -mode frequencies of 59–178 Hz (defaulted-off junction).
- “Converged to 0.2%” and “ f -mode damping agrees to 3.6%” from the spherical BDNK engine (a crossing, documented in `test_sph_shear_qnm.jl`).

B. Defects still present in src/

1. `d2pde2` has no `TabulatedBarotrope` method: the radial solver is blocked on every tabulated EOS.

2. `piecewise_polytrope` ships with $N = 2000$ table nodes: a silent 6% error.
3. `RadialModes` atmosphere floor: `Inf/NaN` or a spurious fundamental depending on table bounds.
4. `GravityModes.jl:324` classifier tie-break; `:276` scan floor above the g -tower for $\delta \lesssim 0.02$.
5. `test_universal_relations.jl:80` omits `Lunit_km=1.0`.
6. Polar operator velocity recovery omits the lapse ($\tau_R = \tau_Q/\alpha$); distinct from the previously fixed lapse bug in `CowlingBDNK3D`.
7. `setup_evo3d` default `cut=false`: -6.95% on f with no symptom.
8. σ_{KO} is an uncontrolled systematic ($\pm 0.25\%$ cut-cell, > 7 pp masked).
9. `Background3D.build_star3d:116` sets $\rho_0 = \varepsilon - p$, exact only for $\Gamma = 2$; harmless to the linear engines, fatal to any nonlinear user.
10. `DGCart3D` background erodes -10% in 4.6 periods (`_limit!` is not well-balanced).
11. `test_frame_independence.jl` block (c) asserts an offset that was the old BC.
12. Julia: `2f1` is the `Float32` literal 20.0.

XI. CONCLUSIONS

The toolkit’s one-dimensional solvers are validated against the published benchmarks they target to 0.02–0.05% (f/p tower), 0.0002% (axial viscous), and literature grade (I-Love), and the BDNK frame machinery reproduces the Clarisse characteristic speeds from (a_1, a_2) alone. On that foundation: hydrodynamic observables are frame-invariant with a live control; a_1 is inert wherever the perturbation carries no scalar content; the relativistic Cowling approximation breaks f -Love and must not be used in an inversion; no two-mode ratio is EOS-independent, but a four-observable pure-spectral relation and a w -Love relation are; and the interface mode at a first-order phase transition is established with a structural rather than fitted null test. The three 3+1D engines are validated where they can be —

the linear Cowling f -mode to $< 0.1\%$ on smooth barotropes, the BDNK viscous damping rate at third-order convergence — and their limits are recorded where they cannot: the metric is frozen, crusted surfaces destabilise the linear engine, the DG background erodes, and time-domain resolution at affordable cost is 4–22% of f . The single number to fix first is `setup_evo3d`’s default.

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The mode census, frame study, universal-relation search and 3+1D validation were executed as adversarially verified agent workflows; every headline number was re-run by an independent verifier, and the verifiers’ retractions are reproduced above.

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